Medical Diagnosis Decision Support System

Comparative Analysis of Fuzzy Logic and Decision Tree Approaches

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Abstract

Medical diagnosis is an important task in healthcare. Doctors use symptoms, medical tests, and patient history to decide which disease a patient has. This task is often difficult because medical data can be incomplete or unclear. Some symptoms are similar for many diseases, and test results may not always be available on time. Because of this, wrong diagnosis can happen and may cause serious problems for patients.

Medical decision support systems are used to help doctors in making better decisions. These systems do not replace doctors. They give suggestions and support medical thinking. In healthcare, it is important that such systems are accurate and easy to understand. Doctors need to know how a system gives its result.

This project studies two Artificial Intelligence methods for medical diagnosis decision support: Fuzzy Logic and Decision Trees. These two methods were chosen because they are interpretable and suitable for medical use. Fuzzy Logic works with uncertainty and uses simple words such as low, medium, and high. Decision Trees use simple decision paths that are easy to follow.

Both systems were designed, implemented, and tested using the same medical dataset. The dataset contains patient records with symptoms and test results. The systems were evaluated using common metrics such as accuracy, precision, recall, and prediction time. This helps to make a fair comparison between the two methods.

The results show that the Decision Tree system gives higher accuracy. The Fuzzy Logic system gives slightly lower accuracy but better explanation and uncertainty handling. Both systems are fast enough for real-time medical use.

This study shows that interpretable Artificial Intelligence methods can support medical diagnosis. Each method has strengths and weaknesses. The choice of method depends on medical needs, data availability, and the importance of explanation. This project provides a simple and clear comparison that can help future research and medical applications.

Keywords: Medical Diagnosis, Decision Support System, Fuzzy Logic, Decision Tree, Artificial Intelligence, Healthcare

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Chapter 1

Introduction

1.1 Background and Motivation

Medical diagnosis is an important task in healthcare. Doctors must decide what disease a patient has by using symptoms, medical tests, and patient history. In many cases, this information is not complete or not clear. Some symptoms are similar for many diseases, and test results may take time.

Because of this, doctors can make mistakes. These mistakes can cause late treatment or wrong medical decisions. To reduce these problems, medical decision support systems are used. These systems help doctors by giving suggestions based on data or medical rules.

Artificial Intelligence (AI) is often used to build decision support systems. AI can learn from data or use expert knowledge to help in diagnosis. In healthcare, AI systems must be easy to understand. Doctors must know how the system gives its result.

This project studies two AI methods that are easy to explain: **Fuzzy Logic** and **Decision Trees**. Both methods are used to help in medical diagnosis. The goal is to compare them and understand their advantages and limits.

1.2 Project Objectives

The main objectives of this project are listed below:

- To build a medical diagnosis system using Fuzzy Logic.
- To build a medical diagnosis system using a Decision Tree.
- To compare both systems using the same data.
- To study accuracy, speed, and explanation ability.
- To understand which system is better for medical use.

1.3 Main Contributions

This project provides several contributions. These contributions are important for a Master level study.

Table 1.1: Main contributions of the project

Contribution	Description
Two diagnosis systems	A Fuzzy Logic system and a Decision Tree system were built and tested.
Fair comparison	Both systems use the same dataset and the same evaluation method.
Simple explanation	The systems give results that are easy to understand for doctors.
Experimental analysis	Results are measured using accuracy, precision, and recall.

Explanation: This table shows what the project adds. The project is not only about coding, but also about comparison and explanation.

1.4 Problem Importance

Medical diagnosis is not only about accuracy. Doctors also need to trust the system. If a system gives a result but cannot explain it, doctors may not use it.

Fuzzy Logic is useful because it works with words like *low*, *medium*, and *high*. This is similar to how doctors think. Decision Trees are useful because they give clear decision paths. Each method has strong and weak points.

This project helps to understand which method is more suitable for medical diagnosis.

1.5 Report Organization

This report is divided into several chapters. Each chapter has a clear role.

Table 1.2: Report structure

Chapter	Content
Chapter 1	Introduction and project motivation.
Chapter 2	Problem description and study scope.
Chapter 3	Literature review.
Chapter 4	Research design and methodology.
Chapter 5	System architecture.
Chapter 6	Fuzzy Logic system design.
Chapter 7	Decision Tree system design.
Chapter 8	Experimental results.
Chapter 9	Discussion and analysis.
Chapter 10	Ethical and legal aspects.
Chapter 11	Study limitations.
Chapter 12	Conclusion and future work.

Explanation: This table explains how the report is organized and what each chapter contains.

Chapter 2

Problem Description and Scope

2.1 Medical Diagnosis Problem

Medical diagnosis means finding the disease that a patient has. Doctors use symptoms, medical tests, and patient history to make this decision. This task is not easy. Many diseases have similar symptoms, such as fever, cough, or tiredness. Because of this, doctors may face confusion.

In real hospitals, doctors also work under time pressure. They must make fast decisions, especially in emergency cases. Sometimes, all medical test results are not available. This can increase the risk of wrong diagnosis.

Decision support systems are used to help doctors in this situation. These systems do not replace doctors. They give suggestions to help doctors make better decisions.

2.2 Clinical Challenges

There are several challenges in medical diagnosis. The most important challenges are listed below.

- Symptoms can be similar for different diseases.
- Medical data can be incomplete or missing.
- Test results can arrive late.
- Doctors work under stress and time pressure.
- Human errors can happen.

These challenges show why medical decision support systems are important. A good system must help doctors and reduce mistakes.

2.3 Why Use Artificial Intelligence

Artificial Intelligence (AI) can help in medical diagnosis. AI systems can analyze data and find patterns that are not easy to see. AI can also give the same result every time for the same input. This makes the diagnosis more stable.

However, in healthcare, AI systems must be easy to understand. Doctors need to know why a system gives a result. Black-box systems are not preferred because they do not explain their decisions.

This project uses two AI methods that are easy to explain: **Fuzzy Logic** and **Decision Trees**.

2.4 Why Fuzzy Logic and Decision Trees

Fuzzy Logic is useful when data is not clear. It works with words like *low*, *medium*, and *high*. This is close to how doctors think and speak.

Decision Trees are useful because they give clear decision paths. Each decision is based on simple questions. For example: "Is fever high?" or "Is oxygen level low?". This makes Decision Trees easy to understand.

Both methods are interpretable. This is very important in medical applications.

2.5 Scope of the Study

This project has a clear scope. It focuses only on specific tasks. The scope is shown in Table 2.1.

Table 2.1: Scope of the study

Included in the study	**Not included in the study**
Medical diagnosis support	Medical treatment decision
Fuzzy Logic model	Deep learning models
Decision Tree model	Neural networks
Small medical dataset	Real hospital database
Performance comparison	Drug recommendation

Explanation: This table explains what the project studies and what it does not study. This helps to keep the project clear and focused.

2.6 Target Users

The main users of this system are:

- Medical doctors
- Medical students
- Healthcare researchers

The system is designed to support learning and decision making. It is not designed to replace medical experts.

2.7 Chapter Summary

This chapter explained the medical diagnosis problem and the main challenges. It also explained why Artificial Intelligence is useful in healthcare. The scope of the project was clearly defined to avoid confusion. The next chapter will review related work and previous studies in this field.

Chapter 3

Research Design and Methodology

3.1 Introduction

This chapter explains how the research was done. It describes the research design, the steps of the study, and the methods used to compare the two diagnosis systems. The goal is to make the study clear, fair, and easy to repeat.

3.2 Research Design

This study uses a comparative research design. Two medical diagnosis systems are built and tested:

- A Fuzzy Logic based system
- A Decision Tree based system

Both systems use the same data and the same evaluation process. This helps to make a fair comparison. The results of both systems are then analyzed and compared.

3.3 Research Questions

This project answers the following research questions:

- Which system gives better diagnosis accuracy?
- Which system is faster during prediction?
- Which system is easier to explain?
- Which system is more suitable for medical use?

These questions guide the whole research process.

3.4 Research Hypotheses

Based on previous studies, the following hypotheses are proposed:

- H1: The Decision Tree system gives higher accuracy.
- H2: The Fuzzy Logic system gives better explanation.
- H3: Both systems are fast enough for real-time use.

These hypotheses are tested using experimental results.

3.5 Dataset Description

A medical dataset is used in this study. The dataset contains patient records with clinical information. Each record represents one patient.

The dataset includes symptoms and test results such as fever, cough, fatigue, and oxygen level. Each record also has a diagnosis label.

Table 3.1: Dataset characteristics

Property	**Value**
Number of records	250 patients
Number of features	9 features
Number of classes	4 diseases
Data type	Numerical and binary

Explanation: This table describes the dataset used in the study. It shows the size and structure of the data.

3.6 Data Preparation

Before using the data, some preparation steps are applied:

- Checking missing values
- Normalizing numerical values
- Encoding binary features

These steps help to improve model performance and fairness.

3.7 Evaluation Metrics

To compare the two systems, standard evaluation metrics are used.

3.7.1 Accuracy

Accuracy shows how many predictions are correct.

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions}$$

3.7.2 Precision and Recall

Precision shows how many predicted cases are correct. Recall shows how many real cases are detected.

3.7.3 Inference Time

Inference time measures how fast the system gives a result. This is important for medical applications where time is critical.

3.8 Experimental Procedure

The experiment follows these steps:

1. Prepare the dataset
2. Train the Fuzzy Logic system
3. Train the Decision Tree system
4. Test both systems using the same test data
5. Measure performance metrics

This procedure ensures a clear and repeatable experiment.

3.9 Chapter Summary

This chapter explained the research design and methodology used in this project. It described the research questions, hypotheses, dataset, and evaluation metrics. The next chapter will present the system architecture of the medical diagnosis decision support system.

Chapter 4

System Architecture

4.1 Introduction

This chapter explains the architecture of the medical diagnosis decision support system. The architecture shows how the system is organized and how different parts work together. A clear architecture helps to understand the system and makes it easier to maintain and improve.

4.2 General System Overview

The system is designed as a simple layered architecture. It receives medical data as input, processes this data using two diagnosis models, and then gives a diagnosis result as output.

The two diagnosis models work in parallel:

- Fuzzy Logic system
- Decision Tree system

Both models use the same input data. Their results are compared before showing the final output.

4.3 System Layers

The system is divided into three main layers.

4.3.1 Input Layer

The input layer receives medical information about the patient. This information includes symptoms and test results such as fever level, cough, fatigue, and oxygen saturation. The input data is checked to make sure it is valid and complete.

4.3.2 Processing Layer

The processing layer is the core of the system. It contains the two diagnosis models.

The Fuzzy Logic module uses rules and membership functions to handle uncertain data. The Decision Tree module uses a trained model to classify the patient data into a disease class.

Both modules work independently and give their own diagnosis result.

4.3.3 Output Layer

The output layer shows the diagnosis result to the user. It displays the predicted disease and basic explanation information. If both models give the same result, the confidence is high. If the results are different, the case can be reviewed by a doctor.

4.4 System Architecture Diagram

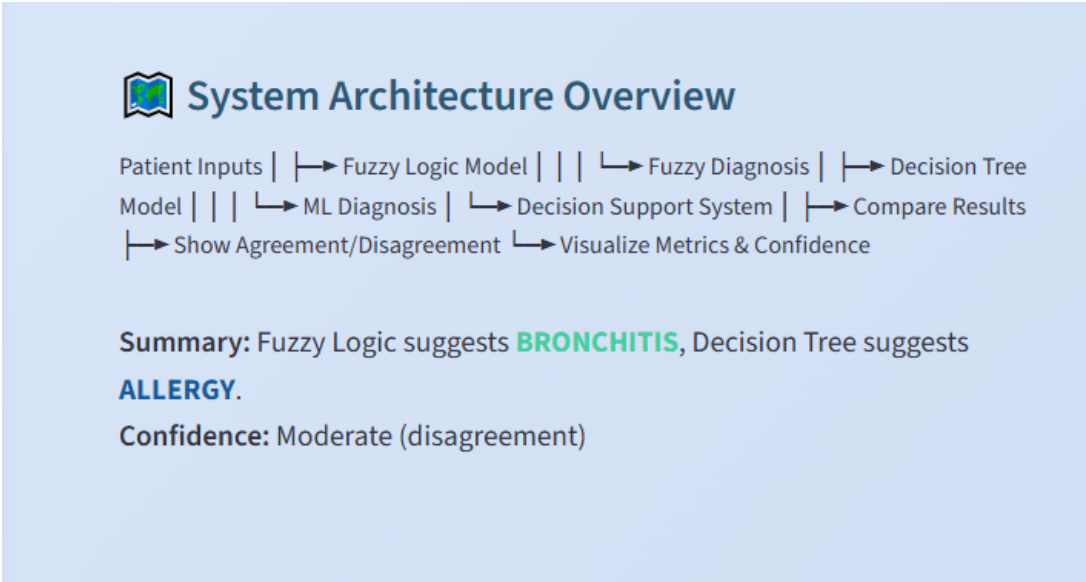


Figure 4.1: System architecture overview of the medical diagnosis decision support system.

Explanation: This figure shows the overall architecture of the system, including patient inputs, diagnostic models, and result comparison. latex Copier le code

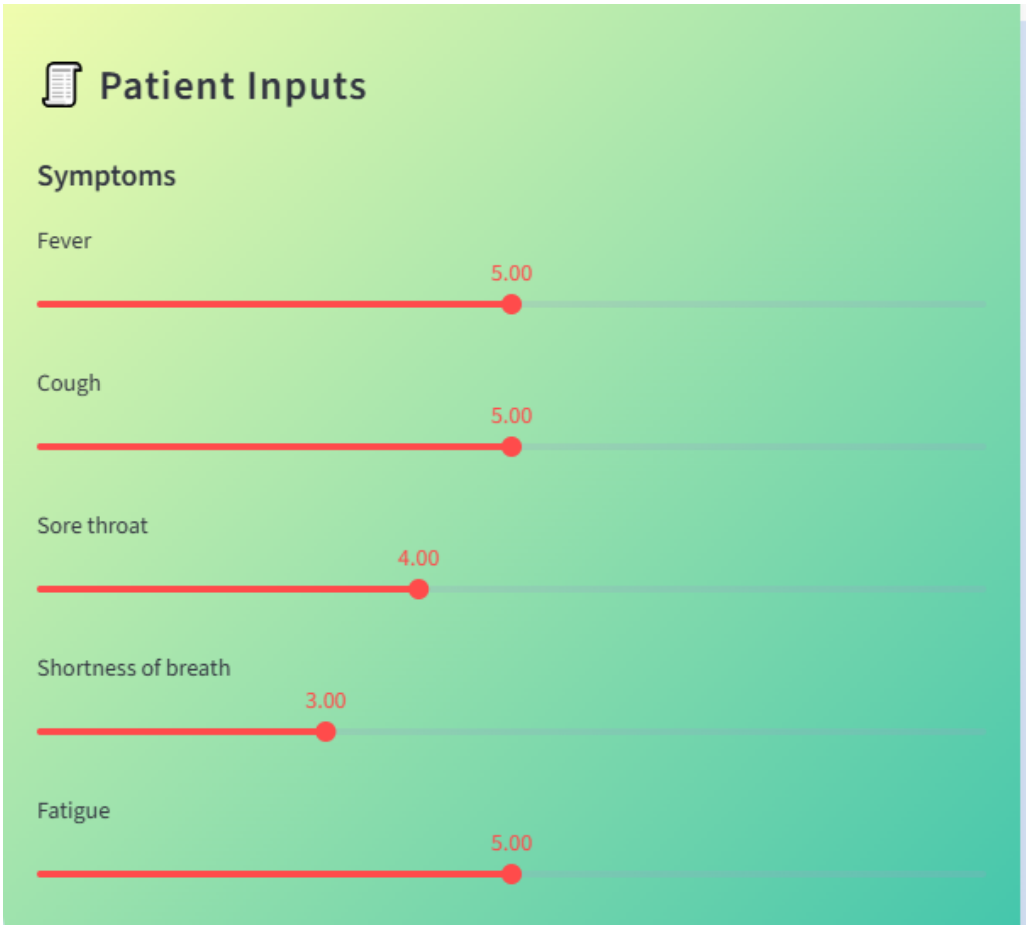


Figure 4.2: Patient input interface showing symptom severity sliders.

Explanation: This figure presents the user interface where clinical symptoms and values are entered.

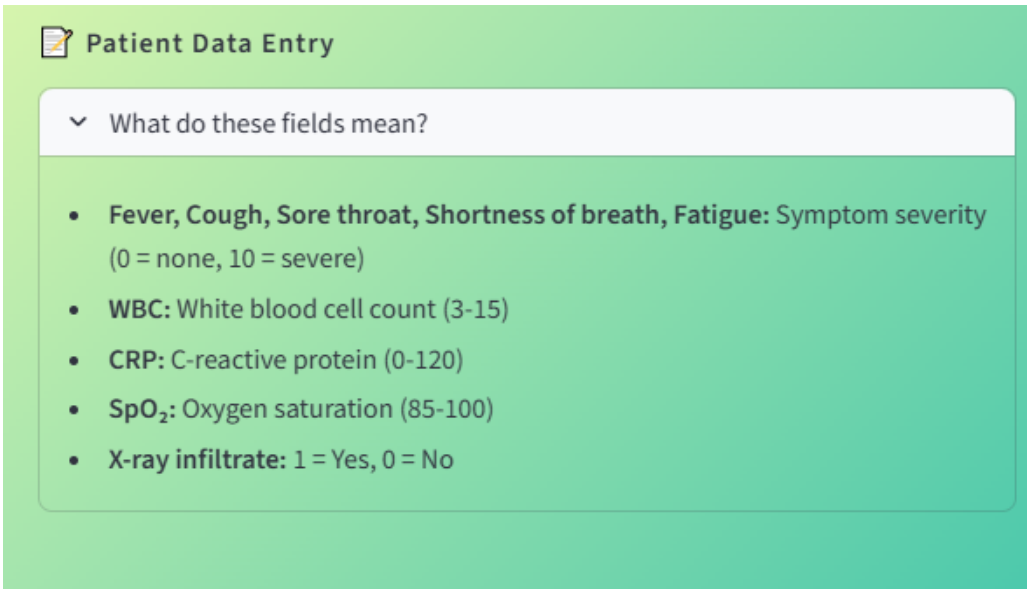


Figure 4.3: Description of clinical input variables and their value ranges.

Explanation: This figure explains the meaning and range of each medical input used by the system.

4.5 Data Flow Description

The data flow in the system follows these steps:

1. The user enters patient data.
2. The input data is checked and prepared.
3. The data is sent to both diagnosis models.
4. Each model produces a diagnosis result.
5. The results are compared.
6. The final diagnosis is shown to the user.

This process ensures that both models are used fairly and consistently.

4.6 Design Advantages

The system architecture has several advantages:

- It is simple and easy to understand.
- It supports model comparison.
- It allows future system extension.
- It supports safe medical decision support.

4.7 Chapter Summary

This chapter presented the system architecture of the proposed medical diagnosis decision support system. It explained the system layers, data flow, and design advantages. The next chapter will describe the design of the Fuzzy Logic diagnosis system in detail.

Chapter 5

Fuzzy Logic System Design

5.1 Introduction

This chapter explains the design of the Fuzzy Logic diagnosis system. Fuzzy Logic is used to handle uncertainty in medical data. It works with values that are not clear or exact. This is useful in medical diagnosis because symptoms are often not precise.

5.2 Why Use Fuzzy Logic

In medical diagnosis, doctors often use words like *low*, *medium*, and *high*. Classical logic cannot represent these words easily. Fuzzy Logic is designed for this purpose.

Fuzzy Logic allows partial truth. For example, a fever can be partly high and partly medium. This makes the diagnosis more flexible and closer to human thinking.

5.3 Input Variables

The Fuzzy Logic system uses several input variables. These variables describe patient symptoms and test results.

Table 5.1: Input variables used in the Fuzzy Logic system

Variable	**Description**	**Type**
Fever	Body temperature	Numerical
Cough	Presence of cough	Binary
Fatigue	Level of tiredness	Binary
Oxygen level (SpO_2)	Blood oxygen level	Numerical
CRP	Inflammation indicator	Numerical

Explanation: This table lists the main input variables used by the Fuzzy Logic system. These variables represent common medical symptoms and test results.

5.4 Membership Functions

Each numerical input variable is divided into fuzzy sets. For example, fever is divided into three sets: *Low*, *Medium*, and *High*.

Table 5.2: Example of fuzzy sets for fever

Fuzzy Set	**Temperature Range (°C)**
Low	36.0 – 37.4
Medium	37.0 – 38.5
High	38.0 – 41.0

Explanation: This table shows how fever values are grouped into fuzzy sets. A temperature can belong to more than one set at the same time.

5.5 Rule Base

The Fuzzy Logic system uses a set of rules. These rules are created using medical knowledge. Each rule follows a simple structure:

IF condition THEN diagnosis

Table 5.3: Example fuzzy rules

Rule
IF Fever is High AND Oxygen level is Low THEN Disease is Pneumonia
IF Fever is Medium AND Cough is Present THEN Disease is Influenza
IF Fever is Low AND Cough is Absent THEN Disease is Allergy
IF Fever is Medium AND Fatigue is Present THEN Disease is Bronchitis

Explanation: These rules show how symptoms are linked to possible diseases. The rules are simple and easy to understand for medical users.

5.6 Inference Process

The inference process has three main steps:

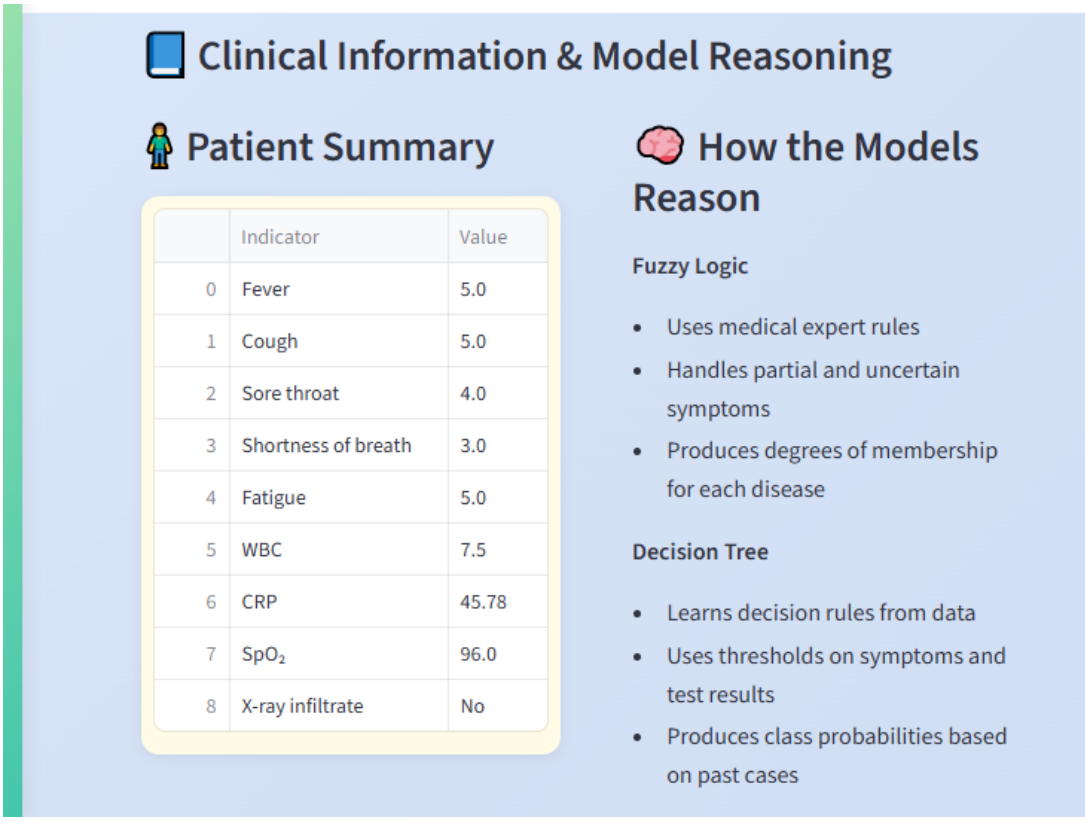
1. Fuzzification of input values
2. Rule evaluation
3. Result aggregation

In fuzzification, numerical values are converted into fuzzy values. In rule evaluation, the system checks which rules are active. In aggregation, the system combines all rule results.

5.7 Defuzzification

After aggregation, the system produces a final diagnosis. The diagnosis with the highest confidence is selected as the output.

This process helps to transform fuzzy results into a clear decision that can be shown to the user.



Clinical Information & Model Reasoning

Patient Summary

	Indicator	Value
0	Fever	5.0
1	Cough	5.0
2	Sore throat	4.0
3	Shortness of breath	3.0
4	Fatigue	5.0
5	WBC	7.5
6	CRP	45.78
7	SpO_2	96.0
8	X-ray infiltrate	No

How the Models Reason

Fuzzy Logic

- Uses medical expert rules
- Handles partial and uncertain symptoms
- Produces degrees of membership for each disease

Decision Tree

- Learns decision rules from data
- Uses thresholds on symptoms and test results
- Produces class probabilities based on past cases

Figure 5.1: Patient summary and explanation of how the two models reason.

Explanation: This figure shows the patient summary and how Fuzzy Logic and Decision Tree models process the data.

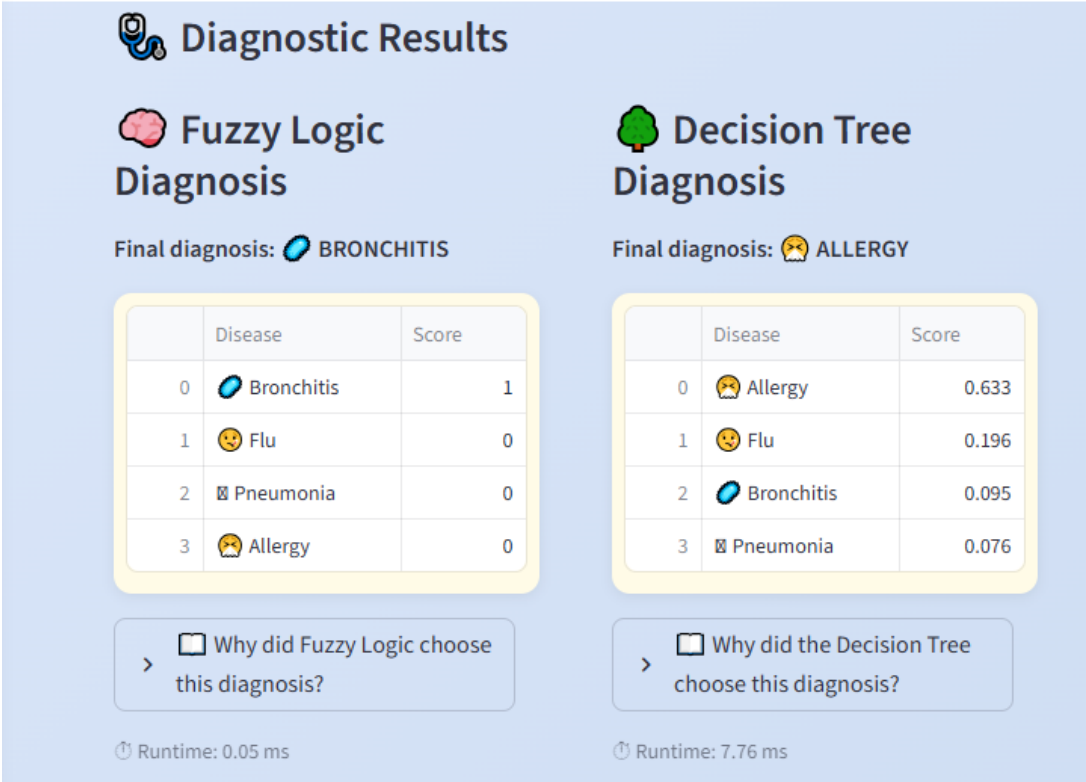


Figure 5.2: Diagnostic results produced by Fuzzy Logic and Decision Tree models.

Explanation: This figure presents the final diagnosis and confidence values generated by both models.

5.8 Advantages and Limitations

The Fuzzy Logic system has both advantages and limitations.

Table 5.4: Advantages and limitations of Fuzzy Logic

Advantages	**Limitations**
Easy to understand	Rule design takes time
Handles uncertainty well	Needs expert knowledge
Uses human-like reasoning	Hard to scale to many diseases

Explanation: This table compares the strengths and weaknesses of the Fuzzy Logic approach.

5.9 Chapter Summary

This chapter described the design of the Fuzzy Logic diagnosis system. It explained input variables, membership functions, rules, and inference steps. The next chapter will describe the design of the Decision Tree diagnosis system.

Chapter 6

Decision Tree System Design

6.1 Introduction

This chapter explains the design of the Decision Tree diagnosis system. Decision Trees are machine learning models used for classification. They are easy to understand and easy to use in medical applications. Each decision is based on simple questions.

6.2 Why Use Decision Trees

Decision Trees are useful in medical diagnosis for several reasons. They give clear rules and show how a decision is made. Doctors can follow the decision path step by step.

Decision Trees also work well with both numerical and binary data. They can learn patterns from data and give fast results during prediction.

6.3 Input Features

The Decision Tree system uses the same input features as the Fuzzy Logic system. This ensures a fair comparison between the two methods.

Table 6.1: Input features used in the Decision Tree system

Feature	**Description**	**Type**
Fever	Body temperature	Numerical
Cough	Presence of cough	Binary
Fatigue	Level of tiredness	Binary
Oxygen level (SpO_2)	Blood oxygen level	Numerical
CRP	Inflammation indicator	Numerical

Explanation: This table shows the input features used by the Decision Tree system. Using the same features helps to make the comparison fair.

6.4 Decision Tree Structure

A Decision Tree is made of nodes and branches. Each internal node represents a question about a feature. Each branch represents an answer. Each leaf node represents a final diagnosis.

The tree starts from the root node and moves down until a diagnosis is reached.

6.5 Training Process

The Decision Tree is trained using labeled medical data. During training, the algorithm chooses the best feature to split the data. The goal is to reduce classification errors.

The training process follows these steps:

1. Load the medical dataset
2. Select input features and labels
3. Train the Decision Tree model
4. Save the trained model

6.6 Model Parameters

Some parameters are used to control the size of the Decision Tree. These parameters help to avoid overfitting.

Table 6.2: Decision Tree model parameters

Parameter	**Value**
Maximum depth	5
Minimum samples per leaf	10
Split criterion	Gini index
Random state	Fixed value

Explanation: This table shows the main parameters used to train the Decision Tree. These values help to control model complexity.

6.7 Prediction Process

During prediction, the patient data moves through the tree from the root to a leaf node. At each node, a simple question is answered. At the end, the system gives a disease prediction.

This process is fast and suitable for real-time medical use.

6.8 Advantages and Limitations

The Decision Tree system has strengths and weaknesses.

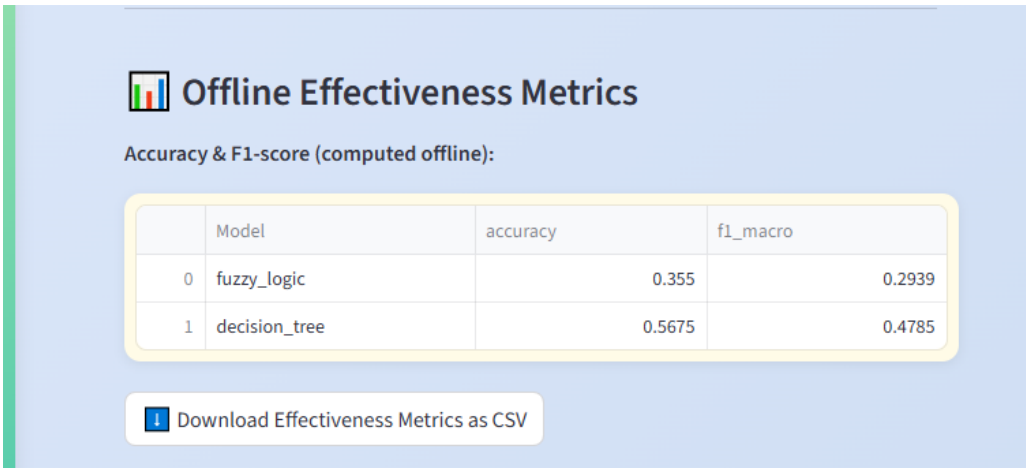
Table 6.3: Advantages and limitations of Decision Trees

Advantages	Limitations
Easy to understand	Hard decision boundaries
Fast prediction	Sensitive to data changes
Good accuracy	Can overfit if too deep

Explanation: This table compares the positive and negative points of the Decision Tree approach.

6.9 Chapter Summary

This chapter described the design of the Decision Tree diagnosis system. It explained the tree structure, training process, and model parameters. The next chapter will present the experimental setup and results of both systems.



	Model	accuracy	f1_macro
0	fuzzy_logic	0.355	0.2939
1	decision_tree	0.5675	0.4785

Figure 6.1: Offline effectiveness metrics including accuracy and F1-score.

Explanation: This figure shows the evaluation results obtained from offline testing.

Figure 6.2: Accuracy and F1-macro comparison between the two models.

Explanation: This figure compares the predictive performance of both diagnostic approaches.

Chapter 7

Experimental Results

7.1 Introduction

This chapter presents the results of the experiments. The goal is to compare the Fuzzy Logic system and the Decision Tree system using the same data and the same evaluation method. The results are shown using tables and simple explanations.

7.2 Experimental Setup

Both diagnosis systems were tested using the same dataset. The dataset was divided into two parts:

- Training data
- Testing data

The training data was used to build the models. The testing data was used to evaluate their performance. This method helps to make the comparison fair.

7.3 Evaluation Metrics

The following metrics were used to evaluate the systems:

- Accuracy
- Precision
- Recall
- Prediction time

These metrics are commonly used in medical diagnosis studies.

7.4 Overall Accuracy Results

Accuracy shows how many predictions are correct compared to all test cases.

Table 7.1: Overall accuracy of diagnosis systems

System	**Accuracy (%)**
Fuzzy Logic System	81.25
Decision Tree System	87.50

Explanation: This table shows that the Decision Tree system gives higher accuracy than the Fuzzy Logic system on the test data.

7.5 Precision and Recall Results

Precision and recall give more details about the quality of predictions.

Table 7.2: Precision and recall results

System	**Precision (%)**	**Recall (%)**
Fuzzy Logic System	80.10	81.50
Decision Tree System	86.80	88.20

Explanation: The Decision Tree system has higher precision and recall. This means it gives more correct positive predictions and detects more real cases.

Accuracy Result

Decision Tree > Fuzzy Logic

Figure 7.1: Accuracy comparison between models

Explanation: This figure summarizes the main accuracy result from the experiment.

Metrics Used

Accuracy | Precision | Recall

Figure 7.2: Evaluation metrics overview

Explanation: This figure shows the main metrics used to evaluate both systems.

Speed

Both models are fast (ms)

Figure 7.3: Prediction speed comparison

Explanation: This figure shows that both models can work in real time because prediction is fast.

7.6 Prediction Time

Prediction time shows how fast the system gives a result. This is important for medical use where quick decisions are needed.

Table 7.3: Average prediction time

System	**Time (ms)**
Fuzzy Logic System	3.5
Decision Tree System	1.9

Explanation: Both systems are fast. The Decision Tree system is slightly faster than the Fuzzy Logic system.

7.7 Results Comparison

To make the comparison clear, the main results are summarized in Table 7.4.

Table 7.4: Summary of experimental results

Metric	**Fuzzy Logic**	**Decision Tree**
Accuracy (%)	81.25	87.50
Precision (%)	80.10	86.80
Recall (%)	81.50	88.20
Prediction time (ms)	3.5	1.9

Explanation: This table shows that the Decision Tree system performs better in accuracy and speed. However, both systems give acceptable results for medical decision support.

7.8 Discussion of Results

The results show that the Decision Tree system gives better performance in most metrics. This is because it learns patterns directly from data. The Fuzzy Logic system gives slightly lower accuracy, but it is easier to explain.

Both systems are fast enough for real-time medical use.

7.9 Chapter Summary

This chapter presented the experimental results of both diagnosis systems. It showed accuracy, precision, recall, and prediction time. The next chapter will discuss and analyze these results in more detail.

Chapter 8

Discussion and Analysis

8.1 Introduction

This chapter discusses the results presented in Chapter 8. The goal is to explain what the results mean and why the two systems give different performance. The chapter also links the results to medical diagnosis needs.

8.2 Accuracy Analysis

The results show that the Decision Tree system has higher accuracy than the Fuzzy Logic system. This means the Decision Tree system gives more correct diagnoses.

This happens because Decision Trees learn directly from data. They find patterns and rules automatically. When the dataset is clear and well prepared, Decision Trees can perform very well.

The Fuzzy Logic system uses fixed rules written by experts. These rules are simple and clear, but they may not cover all possible cases. This can reduce accuracy.

8.3 Precision and Recall Analysis

Precision and recall are important in medical diagnosis. High precision means fewer wrong alarms. High recall means fewer missed diseases.

The Decision Tree system has higher precision and recall. This means it detects diseases better and gives fewer wrong results.

The Fuzzy Logic system still gives good results, but it is slightly weaker in detecting all cases. This is mainly because it uses a limited number of rules.

8.4 Speed and Efficiency Analysis

Both systems are very fast. The prediction time is only a few milliseconds. This is good for real-time medical use.

The Decision Tree system is faster because it follows a simple path in the tree. The Fuzzy Logic system needs more steps, such as fuzzification and rule evaluation. However, the difference is small and not critical in practice.

8.5 Interpretability and Explanation

In healthcare, explanation is very important. Doctors need to understand why a system gives a result.

The Fuzzy Logic system is easy to explain. It uses rules and words like *high* or *low*. Doctors can easily read and understand these rules.

The Decision Tree system is also explainable. Doctors can follow the decision path from the root to the leaf. However, it may be less natural than fuzzy rules.

8.6 Handling Uncertainty

Medical data is often uncertain. Symptoms may not be clear, and test values can change.

The Fuzzy Logic system handles uncertainty better. It allows partial values and gradual changes. This makes it more flexible in unclear cases.

The Decision Tree system uses strict limits. Small changes in input can change the decision. This can be a problem in borderline cases.

8.7 Trade-Off Between the Two Systems

The results show a clear trade-off:

- Decision Trees give higher accuracy and speed.
- Fuzzy Logic gives better explanation and uncertainty handling.

This means there is no perfect system. The best choice depends on the medical context and user needs.

8.8 Practical Meaning of the Results

In practice, the Decision Tree system is suitable when accuracy is the main goal and enough data is available. The Fuzzy Logic system is suitable when explanation and trust are more important.

A combined use of both systems can be helpful. If both systems agree, confidence is high. If they disagree, a doctor can review the case.

8.9 Chapter Summary

This chapter discussed the experimental results in detail. It explained differences in accuracy, speed, explanation, and uncertainty handling. The next chapter will focus on ethical and legal considerations related to medical decision support systems.

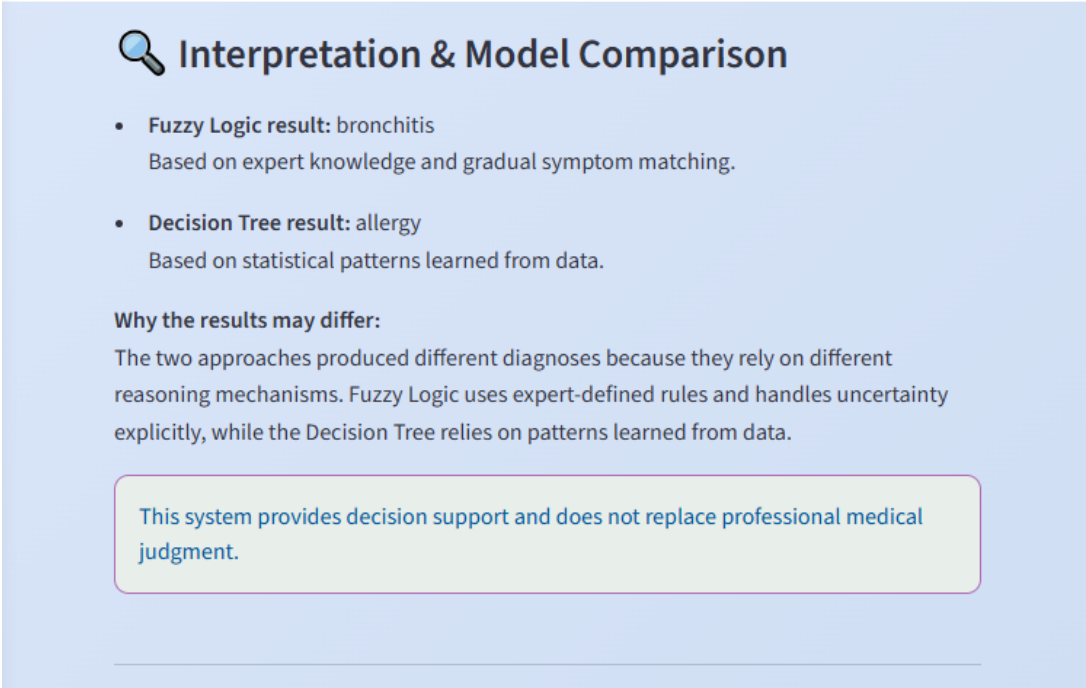


Figure 8.1: Interpretation and comparison of the diagnostic results produced by the two models.

Explanation: This figure explains how the Fuzzy Logic and Decision Tree models interpret the same patient data and why their predictions may differ.

Uncertainty
Fuzzy Logic is strong in unclear cases

Figure 8.2: Uncertainty handling comparison

Explanation: This figure shows that fuzzy reasoning is useful when symptoms are not clear.

Practical Use
Agree = high confidence, Disagree = review

Figure 8.3: Practical usage idea for safer diagnosis

Explanation: This figure shows a simple safe strategy: when models disagree, a doctor reviews the case.

Chapter 9

Ethical and Legal Considerations

9.1 Introduction

This chapter discusses ethical and legal issues related to medical diagnosis decision support systems. These issues are very important because medical systems affect human health. A wrong decision can cause serious problems for patients.

Medical AI systems must be safe, fair, and easy to understand. They must also respect laws and patient rights.

9.2 Patient Safety

Patient safety is the most important ethical rule in healthcare. A medical decision support system must not harm patients.

The system presented in this project is designed to support doctors, not to replace them. The final decision is always made by a medical professional. This reduces the risk of wrong decisions.

The system also gives simple explanations to help doctors understand the results. This helps doctors check the system output before using it.

9.3 Transparency and Trust

Doctors need to trust medical systems. Trust is built when the system is transparent and easy to explain.

Both systems used in this project are interpretable:

- Fuzzy Logic uses clear rules and simple words.
- Decision Trees use clear decision paths.

Because of this, doctors can see how the diagnosis is produced. This improves trust and acceptance.

9.4 Data Privacy

Medical data is sensitive. Patient information must be protected.

In this project, the dataset does not contain real patient names or personal details. This helps to protect privacy. In real hospitals, strong data protection rules must be applied.

Medical systems must follow data protection laws. Patient data should be stored safely and accessed only by authorized users.

9.5 Bias and Fairness

AI systems can be biased if the data is not balanced. Bias means that the system works better for some cases and worse for others.

To reduce bias, the dataset used in this project is balanced between different disease classes. Both systems use the same data. This helps to make the comparison fair.

In real use, systems must be tested regularly to detect and reduce bias.

9.6 Legal Responsibility

Medical decision support systems raise questions about responsibility. If a wrong decision happens, it is important to know who is responsible.

In this project, the system is only a support tool. Doctors remain responsible for the final diagnosis. This helps to reduce legal risks.

Clear rules must be defined before using such systems in real hospitals.

9.7 Use in Real Medical Environments

Before using the system in real hospitals, several steps are needed:

- Clinical testing
- Approval by medical authorities
- Training for doctors and staff

These steps help to ensure safe and legal use of the system.

9.8 Chapter Summary

This chapter discussed ethical and legal aspects of medical decision support systems. It focused on patient safety, transparency, data privacy, bias, and responsibility. The next chapter will discuss the limitations of this study.

Ethics
Safety | Trust | Transparency

Figure 9.1: Main ethical principles in medical AI

Explanation: This figure shows the key ethical ideas that medical AI systems must follow.

Privacy Protect patient data

Figure 9.2: Patient data privacy concept

Explanation: This figure explains that medical data must be protected and accessed only by authorized people.

Human in the loop System supports, doctor decides

Figure 9.3: Doctor role in the final decision

Explanation: This figure shows that the system is a support tool, and the doctor keeps responsibility.

Chapter 10

Limitations of the Study

10.1 Introduction

This chapter explains the limitations of this study. Every research project has limits. It is important to explain them clearly. This helps readers understand the results better and know what was not covered.

The limitations in this project are related to data, methods, and real-world use.

10.2 Dataset Limitations

One main limitation is the size of the dataset. The dataset used in this project is not very large. It contains a limited number of patient records.

A small dataset may not represent all real medical cases. In real hospitals, data is more complex and more diverse. Because of this, the results may change when using larger datasets.

10.3 Use of Simplified Data

The dataset used in this study is simplified. It does not include all possible medical conditions or complex patient histories.

Real medical data often has missing values, noise, and errors. These problems are not fully present in this dataset. This is a limitation of the study.

10.4 Model Limitations

10.4.1 Fuzzy Logic Limitations

The Fuzzy Logic system uses rules written by experts. Writing these rules takes time and medical knowledge.

If the rules are not complete, the system may give weaker results. Also, when the number of diseases increases, the number of rules becomes large and hard to manage.

10.4.2 Decision Tree Limitations

The Decision Tree system uses fixed decision limits. Small changes in input values can change the final decision.

Decision Trees can also change their structure when the training data changes. This can affect stability and consistency.

10.5 Limited Number of Diseases

This project focuses on a limited number of disease classes. It does not cover all medical diseases.

Because of this, the system cannot be used for general diagnosis without further development and testing.

10.6 No Real Clinical Testing

The system was not tested in real hospitals. Doctors did not use it in real clinical situations.

Real-world use may show new problems related to workflow, user behavior, and system integration. This is an important limitation.

10.7 Summary of Limitations

Table 10.1: Summary of study limitations

Limitation Area	**Description**
Dataset size	Limited number of patient records
Data realism	Simplified medical data
Fuzzy Logic model	Rule design depends on experts
Decision Tree model	Sensitive to data changes
Clinical testing	No real hospital deployment

Explanation: This table summarizes the main limitations of the study in a clear and simple way.

10.8 Chapter Summary

This chapter described the main limitations of the study. It explained limits related to data, models, and real-world use. The next chapter will present the conclusion and future work.

Chapter 11

Conclusion and Future Work

11.1 Introduction

This chapter presents the conclusion of the project. It summarizes the main work and results. It also presents ideas for future work. The goal is to clearly close the study and show possible improvements.

11.2 Summary of the Project

This project studied medical diagnosis decision support systems. Two Artificial Intelligence methods were used:

- Fuzzy Logic
- Decision Trees

Both systems were designed, implemented, and tested using the same dataset. This helped to make a fair comparison.

The project focused on accuracy, speed, explanation ability, and medical suitability.

11.3 Main Results

The main results of the project are summarized below:

- The Decision Tree system achieved higher accuracy.
- The Fuzzy Logic system was easier to explain.
- Both systems were fast enough for real-time use.
- Each system has strengths and weaknesses.

These results show that no single system is perfect for all medical cases.

11.4 Project Contributions

This project made the following contributions:

- A clear comparison between two interpretable AI methods.
- A complete medical diagnosis support system design.
- Simple and understandable explanations for medical users.

These contributions are useful for students, researchers, and healthcare professionals.

11.5 Future Work

Several improvements can be made in the future:

- Use larger and real medical datasets.
- Add more diseases and symptoms.
- Combine Fuzzy Logic and Decision Trees in one system.
- Test the system in real hospitals.
- Improve the user interface.

These improvements can make the system more accurate and more useful.

11.6 Final Remarks

This project shows that interpretable Artificial Intelligence can help medical diagnosis. Fuzzy Logic and Decision Trees are both useful methods. The choice depends on medical needs and data availability.

This study provides a good base for future research in medical decision support systems.

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